Help Desk: Third Shift

A CTF for Cisco NetAcad **Security and Connectivity Support** (Modules 1–3) · Game 3 · Facilitator guide & answer key · **Instructor only**

> **At a glance:** 44 challenges · 9,300 points · 11 categories (all start with "SC") following topics 1.1–3.7 · 13 evidence files · loads next to Games 1 and 2 · tested on CTFd 3.8.6 alongside both.

The story

Students are now **Network & Security Support** at Sawgrass Logistics. They work through directory services, MFA, mapped drives, lockouts in a domain controller log, NTFS permissions, and peripherals. Then come Wi-Fi standards, a PoE budget, IPv4/IPv6 addressing, gateways, DHCP and DNS, and reading ipconfig, tracert and netstat output. Last are threat actors, malware, social engineering, a real-looking phishing email, incident response and data classification. In the boss ticket, students follow a phishing campaign from the email through the proxy log to the one compromised PC and the malicious process.

> **Built from your decks:** unlike Game 2, these instructor decks have full slide content. Every answer is tied to a specific slide, cited as "Source: M# S##" in the key (Module and slide number in the instructor PPTX).

Setup

1. Set up CTFd and finish the setup wizard. See `docs/platform-setup.md` in the OmniTech-Arena repository.
2. In CTFd as admin: **Settings** → **Access Tokens** → **Generate**.
3. From the repository root: `pip install requests`, then `python tools/deploy.py tracks/helpdesk-cisco-it-support/03-third-shift_security-and-connectivity-support --url http://<host>:8000 --token ctfd_...`
4. Look for `Done: 44 challenges loaded.` Run it **once**. Other games aren't touched. Add `--hidden` to load the game hidden.
5. One time per server: paste the two sort functions from `docs/platform-setup.md` into **Admin Panel** → **Config** → **Theme** so each game's categories stay grouped.

Running it alongside Canvas

- Each category maps to NetAcad topics (the numbers in parentheses), so students can do "SC 2B" right after finishing topics 2.4–2.5.
- Everything can be solved from the files and the course. No internet or VM is needed.
- Good debriefs: *Verify, Then Trust* (one attempt only, so it's worth discussing why), *Who's Listening?* (what to do when you find a suspicious connection: collect evidence and escalate, don't "fix" it yourself), and *Patient Zero* (why the proxy bypass on one VLAN mattered).

Answer key

Every answer below was submitted by a test student on a CTFd 3.8.6 server that already had Games 1 and 2 loaded, and all 44 were accepted (full score: 9,300). Common wrong answers were checked and rejected, for example the other accounts in the lockout log, port 22 on the PoE switch, and ms-teams.exe in the boss.

Challenge	Answer	Why / discussion	Files
SC 1A Directory Services & Shared Drives (1.1)			
Night Shift 50 pts	`third shift`	Source:	–
Which Directory? 200 pts 4 tries	`CDAB`	Source: M1 S13-14 directory services comparison table	–
MFA Morning 250 pts 4 tries	`1243`	Source: M1 S16 MFA problem/solution table	mfa_tickets.txt
Map It Right 250 pts 4 tries	`\\FS01\Accounting`	User typed 'Acounting' (one c). Trailing backslash and any capitalization are accepted. Source: M1 S20 mapping a shared drive	map_drive_ticket.txt
Policy Refresh 100 pts 4 tries	`gpupdate /force`	gpupdate or gpupdate /force. Source: M1 S21 Force a Group Policy update	–
SC 1B Accounts & Permissions (1.2)			
Built-in Groups 100 pts 3 tries	`Guests`	Source: M1 S24 Users and Groups	–
Permission Puzzle 250 pts 4 tries	`BACD`	Source: M1 S25 Windows permission types table	–
Locked Out 300 pts 4 tries	`wturner,DSP-PC09`	Five 4625s from DSP-PC09, then 4740. gliu and mreyes have failures but were never locked out. Source: M1 S30 AD account lockouts	dc01_security_log.csv
Right Console 150 pts 3 tries	`dsa.msc`	Source: M1 S29 + New Terms (Dsa.msc, Lusrmgr.msc)	–
SC 1C Peripherals (1.3)			
Stuck Spooler 150 pts 3 tries	`Print Spooler`	Source: M1 S36 clearing the print queue	–
Peripheral Triage 300 pts 4 tries	`ECFADB`	Source: M1 S41, S52, S56, S58, S60, S69 problem/solution tables	–
Slow Drive, Fast Port 100 pts 2 tries	`blue`	Source: M1 S67 external drive slow	–
SC 2A Network Types, Wired & Wireless (2.1-2.3)			
Network Sizes 200 pts 4 tries	`FADBEC`	Source: M2 S30-35 network diagrams	–
Cable Choice 200 pts 3 tries	`USF`	Source: M2 S41-44 Ethernet cabling	–
Left Behind 250 pts 3 tries	`WH-PRN-02`	802.11b/g is 2.4 GHz only. The 2018 laptop is dual-band 802.11n, so it's fine. Source: M2 S49-50 802.11 standards table, S59 compatibility issues	warehouse_wifi_change.txt
Power Budget 300 pts 4 tries	`Gi1/0/21`	Ports 1-20 use 10x25.5 + 10x6.5 = 320 W. The 60 W kiosk on port 21 would make it 380 W, over the 370 W budget. The 4 W badge reader on port 22 still fits. Source: M2 S59 PoE power issues	poe_switch_SW-WH-01.txt

Wi-Fi 6 100 pts 3 tries	`802.11ax`	Source: M2 S50	–
SC 2B IP Addressing & Gateways (2.4-2.5)			
Private or Public? 250 pts 3 tries	`CE`	172.32.0.9 is just outside 172.16-172.31. 169.254 is APIPA. Source: M2 S73 private IPv4 table, S91 APIPA	–
Squeeze the IPv6 250 pts 4 tries	`2001:db8::a0:0:0:1`	2001:db8:0:0:a0::1 is also accepted. Source: M2 S68-69 IPv6 compression rules	–
GUA or LLA? 200 pts 3 tries	`LGBLG`	Source: M2 S78-79 GUA starts with 2 or 3, LLA fe80::/10; S83 loopback ::1	–
Gateway Gone Wrong 300 pts 3 tries	`10.20.4.1`	Source: M2 S85-87 default gateway and troubleshooting	ipconfig_DSP-PC11.txt
Local or Remote? 300 pts 2 tries	`192.168.10.70`	/26 = blocks of 64, so .37 is in .0-.63 and .70 is in .64-.127. Only 2 attempts. Source: M2 S66, S84 host forwarding decision with subnet mask	–
SC 2C DNS, DHCP & Firewalls (2.6-2.7)			
169.254 Means... 150 pts 2 tries	`DHCP`	Source: M2 S91-92 APIPA	ipconfig_FRT-KSK1.txt
DORA 150 pts 3 tries	`CDAB`	Source: M2 S94 + New Terms	–
Stale Address 150 pts 3 tries	`ipconfig /flushdns`	Source: M2 S90 DNS caching/TTL, S110	–
Let It Through 150 pts 3 tries	`Windows Defender Firewall`	Source: M2 S104 configuring exceptions	–
SC 2D Network Commands (2.8)			
Read the Config 200 pts 4 tries	`3C-58-C2-9A-17-E4,10.20.1.5`	Source: M2 S108 ipconfig	ipconfig_all_HR-LT03.txt
Where Does It Break? 250 pts 3 tries	`192.0.2.200`	Source: M2 S112 traceroute/tracert	tracert_output.txt
Who's Listening? 300 pts 3 tries	`7316`	Port 4444 to 203.0.113.99. A good discussion lead-in to remote-access Trojans. Source: M2 S116-118 netstat; M3 S33 remote-access Trojan	netstat_ano_ACC-PC05.txt
Modern Linux 100 pts 3 tries	`ss`	Source: M2 S120 iproute2	–
SC 3A Threats, Malware & Attacks (3.1-3.4)			
Name That Threat Actor 250 pts 4 tries	`ADBEFC`	Source: M3 S26-27 types of hackers	–
Malware Lineup 300 pts 4 tries	`CAEBFD`	Source: M3 S32-34 virus, Trojan, and other malware tables	–
Attack Category 200 pts 3 tries	`RASDR`	Source: M3 S36-49 network attack categories	–
SC 3B Social Engineering (3.5)			

Hook, Line & Sinker 300 pts 3 tries	`m365-verify-login.example`	Source: M3 S53 hover over links	phish_email.eml
Return to Sender 200 pts 3 tries	`sawgrass-logistics-support.example`	Source: M3 S52-54 red flags, double-check email sources	phish_email.eml
Social Engineering Bingo 300 pts 4 tries	`AECDFBG`	Source: M3 S42-44 social engineering tables	–
Verify, Then Trust 150 pts 1 tries	`B`	Source: M3 S51 'Amelia Answers an Urgent Call'	–
SC 3C Incident Response & Protecting Data (3.6-3.7)			
Investigation Order 200 pts 3 tries	`CEADB`	Source: M3 S60-61 investigating the incident	–
Who Do You Call? 250 pts 3 tries	`EADBC`	Source: M3 S58-59 incident response stakeholders	–
Sort the Data 250 pts 4 tries	`PICFHL`	Source: M3 S70 categories of sensitive data	–
Know, Have, Are 150 pts 3 tries	`HKAHA`	Source: M3 S67-68 MFA factors	–
Can't Deny It 100 pts 3 tries	`non-repudiation`	Source: M3 S72-73 non-repudiation	–
The Rulebook 150 pts 3 tries	`AUP`	Source: M3 S71 company policies	–
SC 4 Boss Ticket			
Patient Zero 500 pts 3 tries unlocks after: Hook, Line & Sinker, Who's Listening?, Investigation Order	`MailboxRestore_Tool.exe`	Proxy: only WH-PC07/ochen got status 200, and it downloaded MailboxRestore_Tool.exe. The email's Received header gives 198.51.100.77; in netstat that's PID 9032, which tasklist shows is MailboxRestore_Tool.exe. Unlocks after Hook Line & Sinker, Who's Listening?, and Investigation Order. Source: M3 S60-61 investigation steps, M2 S116 netstat, M3 S53 links	phish_email.eml proxy_log_2026-09-29.csv WH-PC07_netstat_tasklist.txt